SMART MANHOLE UTILITIES URBAN IOT

Hardware Pin Interconnect Specification & Engineering Wiring Manual

PROJECT GROUP: P077

SCADA Digital Twin v2.4

STATUS: VERIFIED & TESTED

STUDENT ID	TEAM MEMBER NAME	PROJECT DOMAIN / RESPONSIBILITY
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1. System Architecture & Interconnect Bus

The architecture utilizes a dual-processor distributed topology: an **STM32F103C8T6 (ARM Cortex-M3 @ 72MHz)** executing bare-metal CMSIS firmware at 10Hz non-blocking multi-sensor sampling, feeding formatted JSON packets over full-duplex UART (115200 baud) to a dedicated **ESP8266 NodeMCU V3** wireless gateway. The NodeMCU concurrently streams telemetry over local HTTP/WebSocket (<20ms latency) and publishes 15s cloud records to ThingSpeak.

2. Master Pin-by-Pin Hardware Connection Table

HC-SR04 Ultrasonic Level	VCC GND Trig Echo	**5V Pin** GND **PA1** **PA2 (Divider)**	Power Ground GPIO Out GPIO In	VCC MUST connect to 5V (transducers fail below 4.5V). **Echo requires 1k/2k divider** to step 5V down to 3.33V.
MQ-2 Sensor Methane/Gas	VCC GND A0 (Analog)	**5V Pin** GND **PA0 (ADC1_IN0)**	Power Ground Analog In	Heater requires 5V 150mA. Output is 0-3.3V safe analog read by STM32 12-bit ADC (0 - 4095 counts).
MPU-6050 3-Axis Tilt IMU	VCC GND SCL SDA AD0	**3.3V Pin** GND **PB8** **PB9** GND	Power Ground I2C SCL I2C SDA Address	Shared I2C bus with 16x2 LCD. **AD0 tied to GND locks I2C address to 0x68**. Measures cover tilt 0° to 90°.
16x2 I2C LCD PCF8574 Backpack	VCC GND SCL SDA	**5V Pin** GND **PB8** **PB9**	Power Ground I2C SCL I2C SDA	5V provides crisp contrast and bright LED backlight. I2C address is 0x27. Simultaneous 5-parameter display.
DS18B20 Temp Probe	VCC (Red) GND (Black) Data (Yellow)	**3.3V Pin** GND **PB12**	Power Ground 1-Wire Bus	**4.7kΩ pull-up resistor between Data and 3.3V is MANDATORY**. Non-blocking asynchronous conversion at 10Hz.
Reed Switch Tamper / Anti-Theft	Terminal 1 Terminal 2	**PB13** GND	GPIO In Ground	Configured with internal hardware PULL-UP on PB13. Magnet closed: PB13=0 (SEAL). Magnet removed: PB13=1 (OPEN).
Active Buzzer Acoustic Alarm	Positive (+) Negative (-)	**PB0** GND	GPIO Out Ground	Driven HIGH by STM32 when gas > 2200 ADC or cover is OPEN/TILTED > 15°.
STM32 ↔ ESP Gateway Bridge	PA9 (USART1 TX) GND (STM32)	**Pin D7 (GPIO13 RX)** GND (NodeMCU)	UART Link Common GND	**Common Ground is mandatory**. Full JSON telemetry streamed at 115200 baud every 100ms.

3. Ultrasonic Echo 5V-to-3.3V Voltage Divider Protection

CRITICAL ELECTRICAL NOTICE: The HC-SR04 operates at 5.0V, outputting a 5.0V TTL pulse on its Echo pin. STM32 pin **PA2 is an analog-capable ADC pin and is NOT 5V tolerant (Vdd + 0.3V = 3.6V max)**. Directly connecting 5V Echo will destroy pin PA2. A two-resistor voltage divider is required:

CIRCUIT SCHEMATIC TOPOLOGY	MATHEMATICAL DERIVATION

`[HC-SR04 ECHO (5V)] ■■[ R1 = 1 kΩ ]■■■■■[ STM32 PA2 Input ]`
`■`
`[ R2 = 2 kΩ ]`
`■`
`[ GND ]`
	Formula: Vout = Vin × [R2 / (R1 + R2)]
	Calculation: Vout = 5.0V × [2000Ω / (1000Ω + 2000Ω)]
	Result: Vout = 5.0V × 0.6667 = **3.333 V (100% Safe for STM32 PA2)**

4. Physical 16x2 Character LCD Live Screen Interface

All 5 telemetry parameters are rendered simultaneously in real time without flickering or page switching:

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Row 1: [ D: 25cm G:790 ] (Distance D + Gas ADC G)
Row 2: [ T:28C TL: 0 SEAL ] (Temperature T + Tilt Angle TL + Seal/Open)
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LCD FIELD	PARAMETER	NORMAL STATE	CRITICAL / ALARM TRIGGER
`D: [X]cm`	Ultrasonic Air Gap to Water	50cm down to 10cm	< 5cm (Chamber Flood / Surcharge Alarm)
`G: [X]`	MQ-2 Combustible Gas ADC	400 - 900 ADC	> 2200 ADC (Methane / Sewer Explosive Risk)
`T: [X]C`	Sub-surface Temperature	20°C - 35°C	> 60°C (Underground Cable / Duct Fire)
`TL: [X]`	MPU6050 Live Tilt Angle	0° - 4° (Flat on road)	> 15° (Cover Displaced / Traffic Hazard)
`SEAL / OPEN`	Cover Security Status	SEAL (Magnet attached)	OPEN (Physical Tamper / Cover Removed)

5. Step-by-Step Bench Checkout Protocol

1. Common Ground Rail: Verify with multimeter continuity that STM32 GND and NodeMCU GND are tied together.

2. I2C Diagnostic Splash: On power-up, LCD displays MPU OK (0x68) for 1.5 seconds. If NO MPU SENSOR! appears, verify AD0 is connected to GND.

3. Live Tilt Verification: In normal flat position, LCD displays TL: 0. Tilt the breadboard by 30° to 45°: LCD immediately updates to TL:35 OPEN and buzzer sounds.

4. Ultrasonic Distance Range: Place hand 15cm from HC-SR04: LCD displays D: 15cm. Remove hand: distance smoothly tracks to wall/ceiling without clamping to 0.

5. ThingSpeak & 3D Twin: Telemetry uploads every 15s to ThingSpeak Cloud. Open https://pvkkr.github.io/smart-manhole-twin/: water volume and 3D cover mirror physical sensor positions.